

MAP THE WHOLE UNIT ON ONE SCREEN — 29 SYLLABUS HEADS, 4 BLOCKS

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★ HIGHEST-YIELD IN THIS UNIT

- **Mitosis vs Meiosis** — guaranteed differentiate.
- **Hardy-Weinberg law** — statement, conditions, applications, numerical.
- **Heritability** — definition, methods, values table, uses.
- **Repeatability & MPPA** — often paired with h^2 .
- **Components of variance** — $P = A + D + I + E + G \times E$.
- **Modified Mendelian ratios** — the full table.
- **Linkage & crossing over** — RF, cM, map construction.
- **Chromosomal aberrations** — with livestock examples.
- **Sex-linked vs limited vs influenced** — 3-column table.
- **Forces changing gene frequency** — systematic vs dispersive.

1 HISTORY OF GENETICS — THE TIMELINE THEY ASK

1866 — Gregor Johann Mendel, "Experiments in Plant Hybridisation", *Pisum sativum*, 7 characters → **Father of Genetics**

1900 — Rediscovery by de Vries, Correns & Tschermak

1902 — Sutton & Boveri: **chromosome theory of inheritance**

1906 — Bateson coins the word "GENETICS"; also linkage (with Punnett)

1909 — Johannsen coins **gene**, **genotype**, **phenotype**; pure-line theory

1910–15 — T.H. Morgan: sex linkage & linkage maps in *Drosophila* → **Father of Experimental Genetics**

1927 — Muller: X-ray induced mutation

1941 — Beadle & Tatum: one gene–one enzyme

1944 — Avery, MacLeod & McCarty: **DNA is the genetic material**

1953 — Watson & Crick: DNA double helix

1958 — Meselson & Stahl: **semiconservative replication**

1961 — Jacob & Monod: **lac operon**; Nirenberg: genetic code

1983 — Kary Mullis: **PCR**

1996 — Dolly, first cloned mammal (somatic cell NT, Wilmut)

2003 / 2009 — Human genome / **bovine genome** sequenced

Indian: Fisher, Wright & Haldane = founders of **population genetics**; Lush = **Father of Animal Breeding**

2 MITOSIS VS MEIOSIS ★ SURE QUESTION

Feature	MITOSIS	MEIOSIS
Occurs in	Somatic (body) cells	Germ cells (gonads) only
Divisions	One	Two (I reductional, II equational)
Daughter cells	2	4
Chromosome no.	Same, $2n \rightarrow 2n$ (equational)	Halved , $2n \rightarrow n$ (reductional)
Pairing / synapsis	Absent	Present (zygotene, bivalents)
Crossing over	Absent	Present (pachytene, chiasma)
Prophase	Short, simple	Long, 5 sub-stages
Genetic result	Identical to parent — no variation	Genetically different — creates variation
Function	Growth, repair, replacement	Gamete formation ; keeps species number constant

PROPHASE I — "ZIP, PAIR, CROSS, PULL, END"

Leptotene (thin threads) → Zygotene (**synapsis**, synaptonemal complex) → Pachytene (**crossing over**, tetrad) → Diplotene (**chiasmata** visible, repulsion) → Diakinesis (terminalisation, max condensation). "Lazy Zebras Pull Down Dates."

SOURCES OF VARIATION IN MEIOSIS

① **Crossing over** (pachytene) ② **Independent assortment** of bivalents at metaphase I → 2^n gamete types (cattle 2^{30}) ③ random fertilisation. This is **why** full sibs differ.

3 CHROMOSOME NUMBER, TYPE & KARYOTYPE IN LIVESTOCK AND POULTRY ★ EASY MARKS

Species	2n	Species	2n	Species	2n
Cattle (<i>Bos</i>)	60	Goat	60	Chicken	78
Buffalo (river)	50	Sheep	54	Duck	80
Buffalo (swamp)	48	Pig	38	Turkey	80
Yak	60	Horse	64	Quail	78
Mithun	58	Donkey	62	Pigeon	80
Camel	74	Mule (♂ ass × ♀ horse)	63 — sterile	Dog	78
Rabbit	44	Hinny (♂ horse × ♀ ass)	63	Cat	38

TRAP

Cattle 60 and goat 60 — but cattle autosomes are **all acrocentric** while goat/sheep have some metacentrics. **Mule 63 (odd number)** → homologues cannot pair at meiosis → **sterile**. Cattle × buffalo do not produce viable offspring (60 vs 50).

Karyotype = the complete set of metaphase chromosomes of a somatic cell arranged in descending order of size, in pairs, by centromere position. Its diagrammatic representation = **idiogram**. Total gene content = **genome**.

Type (by centromere)	Position	Shape at anaphase
Metacentric	Exactly median	V
Submetacentric	Sub-median (unequal arms)	L
Acrocentric	Near one end (tiny short arm)	J / rod
Telocentric	Terminal (no short arm)	I / rod

Cattle karyotype (60, XX / 60, XY): **29 pairs of acrocentric autosomes** + sex pair; **X is submetacentric (largest)**, **Y is submetacentric/small**. Buffalo (50): 5 pairs submetacentric + 19 pairs acrocentric + XY. **Poultry**: chromosomes are of two kinds — a few large **macrochromosomes** and many tiny **microchromosomes**; sex chromosomes are **Z and W** (male ZZ, female ZW).

Arm nomenclature: **p** = short arm (petit), **q** = long arm; primary constriction = centromere, secondary constriction = **NOR** (nucleolar organiser region), **satellite** beyond it, **telomere** at the tip.

4 MENDEL'S PRINCIPLES

I. DOMINANCE

one allele masks the other in F_1

II. SEGREGATION

"purity of gametes" — the pair separates at meiosis, each gamete gets ONE

III. INDEPENDENT ASSORTMENT

genes on different chromosome pairs assort independently

Cross	F_2 phenotypic ratio	F_2 genotypic ratio
Monohybrid $Aa \times Aa$	3 : 1	1 : 2 : 1
Dihybrid $AaBb \times AaBb$	9 : 3 : 3 : 1	1:2:1:2:4:2:1:2:1 (9 classes)
Test cross ($F_1 \times$ recessive parent)	Mono 1 : 1 · Di 1 : 1 : 1 : 1	Same as phenotypic
Back cross ($F_1 \times$ either parent)	Depends on parent used	—

Formulae for n heterozygous pairs: gamete types 2^n · genotypes 3^n · phenotypes 2^n · F_2 combinations 4^n . (Dihybrid $n = 2 \rightarrow 4$ gametes, 9 genotypes, 4 phenotypes, 16 squares.)

WHY MENDEL SUCCEEDED

Chose **garden pea** — self-pollinating, pure lines available, short generation, easily hybridised, **7 clear-cut contrasting characters on 7 different chromosome pairs** (no linkage); studied **one/two characters at a time**; kept **numerical records** and applied **mathematics/statistics**; carried crosses to F_2 and F_3 .

TRAP

The **test cross** determines whether a dominant-looking animal is homozygous or heterozygous — the basis of **progeny testing for recessive defects in bulls**. Law of segregation is **universal**; independent assortment **fails when genes are linked**.

5 MODIFIED MENDELIAN RATIOS ★ MEMORISE THE WHOLE TABLE

F_2 Ratio	Phenomenon	Classic livestock / poultry example
1 : 2 : 1	Incomplete dominance (F_1 intermediate)	Andalusian fowl Black $\times$ White $\rightarrow$ Blue ; Palomino horse; Shorthorn
1 : 2 : 1	Codominance (both alleles fully expressed)	Roan Shorthorn cattle (red + white hairs both seen); AB blood group; haemoglobin AS
2 : 1	Dominant lethal — homozygote dies	Yellow mice (A^Y); Dexter cattle $\rightarrow$ "bulldog" calf; Creeper fowl
9 : 7	Complementary (duplicate recessive) genes — both dominants needed	Purple flower in sweet pea; white plumage in some fowl crosses
9 : 3 : 4	Recessive epistasis (supplementary)	Coat colour in mice/dogs — cc masks agouti/black
12 : 3 : 1	Dominant epistasis	Coat colour in dogs (I locus); summer squash
13 : 3	Inhibitory / dominant suppressor gene	White Leghorn — dominant I inhibits colour (C_{-})
15 : 1	Duplicate dominant genes	Feathered shanks in fowl; seed colour in wheat
9 : 6 : 1	Duplicate genes with cumulative effect	Fruit shape; some plumage patterns
9 : 3 : 3 : 1 with new phenotypes	Comb shape in fowl — two gene pairs interacting	Walnut ($R_P_$) 9 : Rose (R_pp) 3 : Pea ($rrP_$) 3 : Single ($rrpp$) 1

THE KEY DISTINCTION

Dominance = interaction between **alleles of the SAME locus**.

Epistasis = interaction between **genes of DIFFERENT loci**, where one gene (**epistatic**) masks the expression of another (**hypostatic**). All modified ratios still total **16** — only the grouping changes.

6 PLEIOTROPY · PENETRANCE · EXPRESSIVITY

- **Pleiotropy** — one gene produces several unrelated phenotypic effects. Ex: **Dexter** gene (short legs + bulldog lethal), **Merle** in dogs (coat + deafness + eye defects), **Manx cat** (taillessness + lethal), **Frizzle fowl** (feather + metabolic rate), dwarfism genes.
- **Penetrance** — the **PERCENTAGE of individuals carrying a genotype that actually show the phenotype**. Complete (100%) or incomplete (e.g. 80% penetrance = 8 of 10 carriers show it). A **population/all-or-none** concept.
- **Expressivity** — the **DEGREE or intensity** to which the phenotype is expressed in an individual that does show it. Variable expressivity: white spotting in Holsteins, polydactyly, roan intensity. An **individual/graded** concept.
- **Why they vary:** modifier genes, genetic background, sex, age, nutrition, temperature and other environmental effects.

REMEMBER

Penetrance = "HOW MANY" (yes/no, %). **Expressivity** = "HOW MUCH" (mild $\rightarrow$ severe). Both together explain why a known carrier may look completely normal.

7 MULTIPLE ALLELES & BLOOD GROUPS

Multiple alleles = three or more alternative forms of a gene at the same locus on homologous chromosomes, arising by repeated mutation. An individual (diploid) can carry **only TWO of them**; a population carries all. They show a definite **dominance series** and always affect the same character.

- **Rabbit coat colour:** C (full agouti) $> c^h$ (chinchilla) $> c^h$ (Himalayan) $> c$ (albino).
- **Human ABO:** $I^A = I^B > i$ — $I^A \& I^B$ codominant, both dominant to i .
- **Cattle blood groups:** **11 systems** — A, B, C, F, J, L, M, S, Z, R'S', T'. The **B system is the most complex known in any animal (> 1000 alleles/phenogroups)**.
- **Applications:** parentage verification & pedigree confirmation in AI programmes; detecting **freemartin/chimerism**; blood transfusion; **genetic markers** for breed characterisation; twin diagnosis (monozygotic vs dizygotic).
- **J substance** in cattle is a soluble antigen adsorbed onto RBC; **haemolytic disease of the newborn** in mule foals and piglets arises from blood-group incompatibility.

8 LETHAL GENES

A **lethal gene** kills the individual carrying it in the appropriate genotype, usually before reproductive age.

Type	Effect
Dominant lethal	Homozygote dies; heterozygote shows a marker phenotype $\rightarrow$ 2 : 1 ratio . Ex: Dexter bulldog calf , yellow mouse, Creeper fowl.
Recessive lethal	Only homozygous recessive dies; carriers normal $\rightarrow$ hidden in the population . Ex: bovine achondroplasia, amputated, muscular hypertrophy, BLAD, CVM, DUMPS in Holstein.
Sub-lethal / semi-lethal	Kills some carriers; survival $< 100\%$.
Conditional lethal	Lethal only in a particular environment.

BREEDING RELEVANCE

Recessive lethals persist because **heterozygous carriers are phenotypically normal**. They surface when **inbreeding** rises or when a popular AI sire is widely used. Detection = **test mating / progeny testing** or, today, **DNA carrier testing**; control = **screen and cull carrier sires**.

9 SEX DETERMINATION

Mechanism	Male	Female	Found in
XX-XY (heterogametic male)	XY (hetero)	XX (homo)	All mammals, <i>Drosophila</i>
ZZ-ZW (heterogametic female)	ZZ (homo)	ZW (hetero)	Birds/poultry, reptiles, some fish, moths
XX-XO	XO	XX	Grasshopper, some insects
Haplo-diploidy	Haploid (unfertilised)	Diploid	Honey bee, wasp, ant
Genic balance (X : A ratio)	0.5	1.0	<i>Drosophila</i> (Bridges)
Environmental (temperature)	—	—	Crocodile, turtle, some fish

THE MAMMALIAN RULE — SAY THIS

In mammals the **Y chromosome is male-determining** because it carries the **SRY gene (Sex-determining Region of Y) = TDF** on its short arm. SRY switches the indifferent gonad into a **testis** → Sertoli cells secrete **AMH/MIS** (regresses Müllerian ducts) and Leydig cells secrete **testosterone** (develops Wolffian ducts). **No SRY → ovary by default**. Hence **XXY is male** (Klinefelter) and **XO is female** (Turner) — unlike *Drosophila*, where the X : A ratio decides.

Intersex conditions in livestock: FREEMARTIN — the sterile female of a heterosexual bovine twin pair; anastomosis of foetal placental vessels lets male cells and hormones cross → **XX/XY blood chimerism**, hypoplastic tubular tract; ~92% of heifers born twin to a bull. Others: **polled intersex in goats** (linked to the polled gene), **XXY (Klinefelter, sterile bull)**, **63,XO Turner in mares**, true and pseudo-hermaphrodites, testicular feminisation.

10 DOSAGE COMPENSATION & LYON HYPOTHESIS

Problem: females have two X chromosomes and males only one — yet the amount of X-gene product is equal in both sexes. The mechanism that equalises it is **dosage compensation**.

Lyon hypothesis (Mary Lyon, 1961): in every somatic cell of a female mammal, **one X is randomly inactivated** early in embryonic life; the inactivation is **random (paternal or maternal), permanent, and clonally inherited** by all descendant cells. The condensed inactive X = **BARR BODY (sex chromatin)**, seen at the nuclear membrane; also visible as the “**drumstick**” in neutrophils.

$$\text{Rule Number of Barr bodies} = (\text{Number of X chromosomes}) - 1$$

$$XX = 1 \cdot XY = 0 \cdot XO \text{ (Turner)} = 0 \cdot XXY \text{ (Klinefelter)} = 1 \cdot XXX = 2.$$

THE CLASSIC PROOF

Tortoiseshell / calico cats are almost always female. The orange (O) and black (o) alleles are **X-linked**; random X-inactivation in different patches of skin gives a **mosaic** coat. A **male tortoiseshell is rare and sterile** — he is **XXY**. This single example proves both X-linkage and random X-inactivation.

Note: in **birds (ZZ/ZW)** dosage compensation is incomplete. Barr-body counting was the classical **sexing method** before molecular tests.

11 SEX-LINKED VS SEX-LIMITED VS SEX-INFLUENCED INHERITANCE ★ CERTAIN TABLE QUESTION

Feature	SEX-LINKED	SEX-LIMITED	SEX-INFLUENCED (sex-conditioned)
Gene located on	Sex chromosome (X or Z)	Autosome (may be any chromosome)	Autosome
Expression	According to the sex chromosome carried	Expressed in ONE sex only ; totally absent in the other though genes are present	Expressed in both sexes, but dominance is REVERSED between them
Controlled by	Chromosomal location	Sex hormones / sex-specific organs	Sex-hormone level modifying dominance
Examples	Barred plumage & silver/gold in fowl (Z-linked) , slow feathering; haemophilia, red-green colour blindness, X-linked muscular dystrophy	Milk yield & lactation traits (females), egg production, broodiness; semen quality, scrotal circumference (males); antlers in deer, cock feathering	Horns in sheep (Dorset) — dominant in ♂, recessive in ♀; mahogany & white in Ayrshire cattle ; pattern baldness in man
Key genetic point	Criss-cross inheritance: father → daughter, mother → son. Reciprocal crosses give different results. Male mammals are hemizygous for X genes	Heritability is normal; the trait can only be improved through the opposite sex by progeny testing	Same genotype (heterozygote) gives different phenotype in the two sexes

APPLIED GOLD — AUTO-SEXING OF CHICKS

In poultry the male is **ZZ** and the female **ZW**, and **silver (S) is dominant to gold (s)**, both Z-linked. Mate a **gold male (Z^sZ^s) × silver female (Z^SW)**: every **daughter is GOLD** (her single Z comes from the father) and every **son is SILVER** (Z^SZ^s). This **criss-cross** pattern lets day-old chicks be **sexed by down colour**, saving hatchery labour. The same logic is used with the **barring (B)** gene and with slow/fast feathering.

TRAP

Milk yield is **sex-limited, NOT sex-linked** — the genes are autosomal and the bull carries them fully, which is exactly **why a bull is evaluated on his daughters' records (progeny testing)**. Horns in sheep are **sex-influenced, NOT sex-limited** — both sexes can be horned, only the dominance flips.

12 LINKAGE & CROSSING OVER

Linkage = the tendency of genes located on the **same chromosome** to be inherited **together**, i.e. they do not assort independently. Discovered by **Bateson & Punnett (1906, sweet pea)**; explained by **T.H. Morgan (*Drosophila*)** — the **chromosome theory of linkage**. All genes on one chromosome form a **linkage group**; number of linkage groups = **haploid number (n)** — cattle 30, buffalo 25, chicken 39.

Term	Meaning
Coupling (cis)	Both dominants on one chromosome: AB/ab
Repulsion (trans)	One dominant on each: Ab/aB
Complete linkage	No crossing over; only parental types (♂ <i>Drosophila</i> , ♀ silkworm)
Incomplete linkage	Parental types in excess + a few recombinants — the usual situation

Crossing over = reciprocal exchange of segments between **non-sister chromatids of homologous chromosomes** at **PACHYTENE of prophase I**; the visible point of exchange is the **CHIASMA** (seen at diplotene). Types: single, double, multiple. Factors affecting it: **distance between genes (↑ distance ↑ CO)**, **sex, age, temperature, X-rays and chemicals, proximity to the centromere (reduced)**.

SIGNIFICANCE

Crossing over is the **chief source of genetic recombination** — it creates new gene combinations, allows **linkage mapping**, breaks unfavourable linkages, and underlies **marker-assisted selection**.

13 LINKAGE MAP — RECOMBINATION FREQUENCY

$$\text{RF Recombination frequency \%} = (\text{No. of recombinant progeny} / \text{Total progeny}) \times 100$$

$$\text{Map distance } 1\% \text{ RF} = 1 \text{ map unit (m.u.)} = 1 \text{ centiMorgan (cM)}$$

- **Maximum RF = 50%.** At 50% the genes behave as if **unlinked** (independent assortment) — either on different chromosomes or very far apart on the same one.
- **Recombination frequency is directly proportional to the distance between genes** (Sturtevant, 1913) — this is what makes mapping possible.
- **Linkage maps are constructed from a THREE-POINT TEST CROSS:** the two **rarest (double-crossover) classes** identify the **MIDDLE gene**; the two most frequent classes are parental.
- **Strength of linkage** $\propto 1/\text{distance}$. Closely linked genes → few recombinants.

$$\text{Coincidence } \text{C.o.C.} = \text{Observed double CO} / \text{Expected double CO}$$

$$\text{Interference } I = 1 - \text{C.o.C.}$$

Interference = one crossover reduces the chance of a second nearby crossover (positive interference is the rule). Expected DCO = product of the two individual RFs. **Map types: genetic/linkage map** (distance in cM, from RF) vs **physical map** (distance in base pairs, from sequencing) — the two do not always agree because CO frequency varies along the chromosome.

14 MUTATION

Mutation = a sudden, heritable change in the genetic material not due to segregation or recombination. Term coined by **Hugo de Vries (1901)**.

Mutation is the ULTIMATE/original source of all genetic variation — every allele in every breed originated as a mutation.

By cell type	Somatic — body cell; not inherited ; gives a mosaic/chimera	Germinal — gametic; transmitted to offspring
By origin	Spontaneous — natural, rate 10^{-5} – 10^{-6} /gene/generation	Induced — by mutagens (Muller, X-rays, 1927)
By size	Gene (point) mutation — within one gene	Chromosomal mutation — aberration (see right)

Point mutation	Change & consequence
Substitution	Transition (purine→purine or pyrimidine→pyrimidine) · Transversion (purine→pyrimidine)
Silent (same-sense)	Codon changes but the same amino acid — code is degenerate
Missense	Different amino acid — e.g. sickle-cell haemoglobin
Nonsense	Creates a STOP codon → truncated, non-functional protein
Frameshift	Insertion or deletion of bases (not a multiple of 3) → whole reading frame shifts ; most damaging

MUTAGENS — ① **Physical**: X-rays, gamma rays, UV (forms **thymine dimers**), neutrons, high temperature. ② **Chemical**: base analogues (**5-bromouracil**, **2-aminopurine**) · deaminating agents (**nitrous acid**) · alkylating agents (**EMS**, **mustard gas**) · **acridine dyes** (**frameshift**) · hydroxylamine. ③ **Biological**: transposons ("jumping genes", McClintock), viruses.

Significance: raw material for selection & new breeds; but most mutations are **recessive and deleterious** → genetic defects (BLAD, CVM, DUMPS, achondroplasia); basis of genetic load. Detected by **test mating**, **Muller's CIB technique**, **DNA testing**.

15 CHROMOSOMAL ABERRATIONS ★ HIGH-YIELD

A • NUMERICAL (change in chromosome NUMBER) — chief cause is **NON-DISJUNCTION** (failure of homologues/chromatids to separate) or anaphase lag.

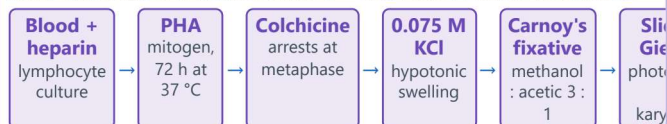
Type	Formula	Example
Euploidy — whole sets	Monoploid (n), triploid (3n) , tetraploid (4n) ; auto- & allo-polyploidy	Rare and usually lethal in mammals; common in plants; seen in early bovine embryonic death
Nullisomy	$2n - 2$ (a whole pair lost)	Lethal
Monosomy	$2n - 1$	63,XO — Turner's in the mare (gonadal dysgenesis, infertile)
Trisomy	$2n + 1$	XXY Klinefelter — sterile bull/boar with small testes; autosomal trisomy in calves (brachygnathia, early death)
Tetrasomy	$2n + 2$	Rare

B • STRUCTURAL (change in chromosome STRUCTURE) — follows breakage and faulty reunion.

Type	What happens & livestock significance
Deletion / deficiency	A segment is lost (terminal or interstitial); pseudo-dominance of recessives; usually harmful. Cri-du-chat in man.
Duplication	A segment is repeated ; source of new genes; less harmful than deletion.
Inversion	A segment breaks, rotates 180° and rejoins. Paracentric (centromere outside) or pericentric (includes centromere). Forms an inversion loop at pairing and acts as a crossover suppressor .
Translocation	Segment shifts to a non-homologous chromosome. Reciprocal (exchange) or ROBERTSONIAN (centric fusion) — two acrocentrics fuse at the centromere, so 2n falls by one . ★ 1/29 Robertsonian translocation in cattle (first found in Swedish Red & White; also in Indian breeds) → carriers are phenotypically normal but show reduced fertility (~5–10% lower conception) , repeat breeding and early embryonic death . All AI bulls must be karyotyped and carriers culled .
Isochromosome • Ring chromosome	Misdivision of centromere; both ends fuse.

16 CYTOGENETICS — TECHNIQUE, BANDING & APPLICATION

Cytogenetics = the branch that studies chromosomes — their structure, number, behaviour and abnormalities — and relates them to inheritance.



Band	Stain / target
G-band	Trypsin + Giemsa ; AT-rich regions dark. Most widely used routine banding .
Q-band	Quinacrine mustard, fluorescence; same pattern as G.
R-band	Reverse of G; GC-rich; good for chromosome ends.
C-band	Constitutive heterochromatin at the centromere ; identifies Y and centric fusions.
NOR / Ag-band	Silver staining of the nucleolar organiser region.
FISH	Fluorescent probe to a specific DNA sequence — modern gene/aberration mapping.

Applications in veterinary practice: screening **AI bulls for 1/29 translocation** before entry into semen stations · diagnosis of **freemartinism (XX/XY chimerism)** and intersex · investigating **repeat breeding, infertility and habitual abortion** · sexing · species & hybrid identification (mule, hybrids) · confirming chromosome number in new/exotic species · cancer cytogenetics.

17 EXTRA-CHROMOSOMAL (CYTOPLASMIC) INHERITANCE

Inheritance controlled by genes located **outside the nucleus** — in **mitochondria** (and plastids in plants) — hence also called **cytoplasmic, maternal or non-Mendelian inheritance**.

- **CRITERIA (how you recognise it)**: ① **Reciprocal crosses give DIFFERENT results** ② the phenotype always follows the **FEMALE (maternal) parent** ③ **no Mendelian segregation ratios** in F_2 ④ the trait **cannot be mapped** to any chromosome ⑤ nuclear substitution/back-crossing does not change it.
- **Why maternal?** The **ovum contributes virtually all the cytoplasm** (and therefore all the mitochondria); the sperm contributes essentially only its nucleus, and the few paternal mitochondria are destroyed.
- **Mitochondrial DNA (mtDNA)**: **circular, double-stranded, ~16.3 kb**, naked (no histones), **37 genes** (13 proteins of oxidative phosphorylation + 22 tRNA + 2 rRNA), **high mutation rate**, many copies per cell, **heteroplasmy** possible.
- **Examples**: milk-yield and growth **maternal/cytoplasmic effects** in cattle · mtDNA haplotypes used for **breed origin, phylogeny and biodiversity studies** · **Leber's hereditary optic neuropathy**, mitochondrial myopathies · **kappa particles** in *Paramecium* · **shell coiling in Limnaea** and pigment in *Ephesia* (maternal-effect genes) · male sterility in plants.
- **Distinguish: maternal EFFECT** = nuclear genes of the **mother** acting through the egg cytoplasm (e.g. dam's milk on calf growth) — it is still Mendelian; **cytoplasmic inheritance** = genes actually located in the cytoplasm.

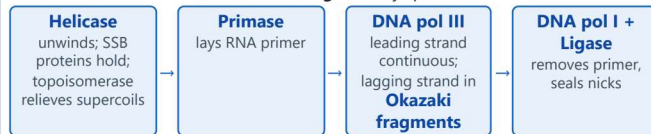
18 NUCLEIC ACIDS • GENE CONCEPT • REPLICATION

DNA (Watson & Crick, 1953): double helix, right-handed (B-form), antiparallel (5'→3' and 3'→5'), sugar-phosphate backbone outside, bases inside. **10 bp per turn**, pitch 3.4 nm, rise 0.34 nm, diameter 2 nm. Base pairing: **A = T (2 H bonds)**, **G = C (3 H bonds)**. **Chargaff's rules:** A = T, G = C, purines = pyrimidines. Building block = **nucleotide** (base + deoxyribose + phosphate); base + sugar = nucleoside.

	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A T G C	A U G C
Strands	Double	Usually single
Function	Stores genetic information; stable	Expresses it (mRNA carries, tRNA transfers, rRNA builds); labile

Gene concept: the **unit of heredity** — a DNA segment coding for one polypeptide or functional RNA. **Beadle & Tatum: one gene—one enzyme → one gene—one polypeptide.** Benzer: **cistron** (function), **recon**, **muton**. Eukaryotic genes are **split** — **exons** interrupted by **introns** removed by **splicing**. Regulation: **lac operon** (Jacob & Monod, 1961).

DNA REPLICATION — **semiconservative** (proved by Meselson & Stahl, 1958, ¹⁵N density-gradient in *E. coli*): each daughter molecule has **one parental + one new** strand. It is **bidirectional**, starts at an **origin**, always proceeds 5' → 3'.



Central dogma: DNA —(transcription)→ mRNA —(translation)→ protein. **Genetic code:** triplet, **64 codons**, **degenerate**, non-overlapping, comma-less, **universal**; **start AUG**, **stop UAA, UAG, UGA**. Reverse transcription in retroviruses.

19 INTRODUCTION TO MOLECULAR TECHNIQUES

Technique	Principle & use in animal genetics
PCR (Kary Mullis, 1983)	In-vitro exponential amplification (2ⁿ) of a target DNA using two primers + Taq polymerase + dNTPs + Mg²⁺ . Three steps repeated 25–40 cycles: DENATURATION 94 °C → ANNEALING 50–65 °C → EXTENSION 72 °C . Variants: RT-PCR, real-time/qPCR, multiplex, nested. Uses: sex determination of embryos, disease-gene & carrier testing (BLAD, CVM), parentage.
Gel electrophoresis	DNA is negatively charged → migrates to the anode ; separation by size in agarose; visualised with ethidium bromide under UV.
RFLP	Restriction enzymes cut DNA at specific sites; a mutation creates/destroys a site → different fragment lengths . Codominant marker; first DNA marker used for mapping.
RAPD • AFLP	Arbitrary/selective primers; quick breed characterisation & biodiversity studies (RAPD is dominant, less reproducible).
Microsatellites (SSR/STR)	Short tandem repeats, highly polymorphic, codominant — the FAO-recommended markers for parentage verification and genetic-distance/AnGR studies.
SNP & SNP chips	Single base differences; hundreds of thousands genotyped at once → the basis of GENOMIC SELECTION (Unit 3).
DNA sequencing	Sanger dideoxy (chain-termination) method; now NGS — whole-genome sequencing.
Blotting	Southern = DNA, Northern = RNA, Western = protein. ("SNoW dRoP")
DNA fingerprinting (Alec Jeffreys)	VNTR/minisatellite profile unique to an individual — parentage, identity, forensic, theft cases.

20 POPULATION GENETICS — GENE FREQUENCY & THE HARDY–WEINBERG LAW ★★ CERTAIN QUESTION

Population genetics studies the **genetic composition of a Mendelian population** — a group of interbreeding individuals sharing a common **GENE POOL** — and how it changes. **Individual genetics asks "what genotype?"; population genetics asks "at what frequency?"** Animal breeding is **applied population genetics**: we change gene frequencies in a herd, not in one cow.

Genotypic freq. D (AA) + H (Aa) + R (aa) = 1
Gene (allele) freq. p = D + ½H ; q = R + ½H ; p + q = 1
By gene counting p = (2×AA + Aa) / 2N
HARDY–WEINBERG p² (AA) + 2pq (Aa) + q² (aa) = 1

Statement (Hardy & Weinberg, 1908, independently): "In a **large, random-mating population** free from mutation, migration, selection and drift, the **gene and genotypic frequencies remain CONSTANT generation after generation.**" Such a population is in **genetic equilibrium**.

SIX CONDITIONS: ① **Large population** (infinitely large) ② **Random mating** (panmixis) ③ **No mutation** ④ **No migration** (no gene flow) ⑤ **No selection** — all genotypes equally fertile & viable ⑥ **No genetic drift**; plus normal meiosis, diploid, sexual, non-overlapping generations.

- **Equilibrium is reached in ONE generation of random mating** for an autosomal locus, whatever the starting genotypic frequencies. For a **sex-linked** locus it is approached gradually, the frequency **oscillating** and halving the difference each generation.
- **Maximum heterozygosity 2pq = 0.5**, occurring when **p = q = 0.5**.
- When q is small, **most recessive alleles are hidden in heterozygotes**: at q = 0.01, carriers (2pq ≈ 0.0198) are **≈ 198 times** as frequent as affected homozygotes (q² = 0.0001). **This is why culling only affected animals removes a recessive defect so slowly.**
- **Testing for equilibrium:** compute p and q from the data, get expected numbers (Np², N2pq, Nq²), apply **χ² = Σ(O–E)²/E** with **df = no. of genotypes – no. of alleles = 3 – 2 = 1** (compare 3.84).
- **Applications:** estimating **carrier frequency** of recessive lethals and of blood-group/coat-colour alleles; predicting genotype frequencies in a herd; measuring genetic variability; a **baseline against which the effect of selection, drift or inbreeding is measured**; forensic and parentage probabilities.

TRAP

Inbreeding changes GENOTYPIC frequencies but NOT gene frequencies — it only redistributes: **AA = p² + pqF, Aa = 2pq(1 – F), aa = q² + pqF**. **Only selection, mutation, migration and drift change gene frequency.** Also: a population showing HWE proportions is **not proof** of no evolution — the test has low power.

21 FORCES CHANGING GENE & GENOTYPIC FREQUENCIES

Force	Formula	Nature & breeding relevance
MUTATION	$\Delta q = up - vq$ at equilibrium $q = u/(u+v)$	Systematic ; direction & amount predictable. Very slow (10 ^{–5} –10 ^{–6}), so negligible in a breeder's lifetime , but the only original source of new alleles .
MIGRATION (gene flow)	$q_1 = m(q_m - q_0) + q_0$ $\Delta q = m(q_m - q_0)$	Systematic . m = proportion of immigrants. Exactly what GRADING UP and exotic semen do — a rapid, controllable way to change gene frequency; erodes indigenous genes if unregulated.
SELECTION	$\Delta q = -spq^2 / (1 - sq^2)$ fitness $w = 1 - s$	Systematic and the breeder's main tool . s = selection coefficient (0 = no selection, 1 = lethal). Natural vs artificial selection. Against a recessive , progress is fast at high q but extremely slow at low q (alleles hide in heterozygotes). Selection against a dominant can eliminate it in one generation.
GENETIC DRIFT (Sewall Wright effect)	$\sigma^2_q = pq / 2N$	DISPERSIVE — magnitude predictable but direction random . Random change in gene frequency due to sampling of gametes in a SMALL population . Leads to random fixation (q=1) or loss (q=0) , ↑ homozygosity, ↓ variability, differentiation between herds. Includes the FOUNDER EFFECT (new herd started from few animals) and BOTTLENECK (population crash). Critical for small closed herds, zoos and endangered breeds.

THE ONE-LINE CLASSIFICATION THEY WANT

SYSTEMATIC forces — **mutation, migration, selection**: both the **amount and the direction** of change are predictable. **DISPERSIVE force** — **random genetic drift**: only the **amount** (variance) is predictable, the **direction is random**. Together with **inbreeding** (which alters genotypic, not gene, frequency) these are the complete set of forces acting on a gene pool.

PUT THE FOUR TOGETHER IN A HERD

A closed Sahiwal herd of 40 cows: **drift** randomly loses rare alleles; **inbreeding** rises, exposing recessive defects; **selection** for milk yield shifts the favourable alleles up; introducing outside bulls = **migration**, which restores variability and can rapidly change frequencies; **mutation** quietly supplies the raw material. **Effective population size (N_e) is the number that decides how strong drift and inbreeding will be** — the link into Unit 3 conservation.

22 QUALITATIVE VS QUANTITATIVE + AVERAGE EFFECT & BREEDING VALUE

	QUALITATIVE	QUANTITATIVE (metric)
Genes	One or few, with large effect (oligogenes)	Many genes, each with small effect (polygenes / minor genes)
Expression	Discontinuous, distinct classes	Continuous, no clear classes
Measured by	Counting / classifying (kind)	Measuring (degree) — kg, litres, cm, number
Environment	Little or no effect	Strongly affected
Distribution	Ratios (3:1, 9:3:3:1)	Normal curve
Examples	Coat colour, horn/poll, blood groups, genetic defects	Milk yield, fat %, growth rate, body weight, egg number, litter size, wool yield

TRAP

Almost every trait of economic importance is **QUANTITATIVE** — which is exactly why animal breeding needs statistics (Unit 1) rather than Punnett squares. **Threshold traits** (twinning, dystocia, disease resistance) look qualitative but have an underlying continuous liability.

Genotypic values (Falconer): midpoint **m** between the two homozygotes; **AA** = +a, **aa** = -a, **Aa** = d. d = 0 no dominance • d = a complete dominance • d > a overdominance.

Population mean $M = a(p - q) + 2pqd$

Average effect of gene substitution $\alpha = a + d(q - p)$

Breeding value **BV** = sum of the average effects of the genes an animal carries

AA = $2q\alpha$ • **Aa** = $(q - p)\alpha$ • **aa** = $-2p\alpha$

Why BV matters: a parent passes on **genes, not genotypes** — dominance and epistatic combinations are broken up at meiosis. So only the **additive** part is transmitted. **BV** = **twice the mean deviation of an animal's progeny** from the herd mean; the estimated BV is what we now call the **EBV**, and it is the value on which **all modern selection (BLUP, genomic selection)** is based. **Transmitting ability** = $\frac{1}{2}$ BV.

24 HERITABILITY (h^2) ★★ + REPEATABILITY (R)

Narrow sense $h^2 = V_A / V_p$ — the one always meant in animal breeding

Broad sense $H^2 = V_G / V_p = (V_A + V_D + V_I) / V_p$

Repeatability $r = (V_G + V_{Ep}) / V_p \rightarrow r \geq h^2$ always

h^2 = the **proportion of the total phenotypic variance that is due to additive genetic (transmissible) effects**; equivalently the **regression of breeding value on phenotype**. It is a **property of a trait IN A POPULATION IN AN ENVIRONMENT**, not of an individual; ranges **0 to 1**; it is **not a constant**.

Method of estimating h^2	Formula
Regression: offspring on ONE parent	$h^2 = 2b$
Regression: offspring on MID-PARENT	$h^2 = b$
Paternal HALF-SIB correlation (best — free of maternal & dominance effects)	$h^2 = 4t$ (t = intraclass correlation)
FULL-SIB correlation (over-estimates: contains $\frac{1}{4}V_D$ + common environment)	$h^2 = 2t$
Realised heritability (from an actual selection experiment)	$h^2 = R / S$ (response ÷ selection differential)

h^2 class	Typical traits (memorise the pattern)	Best selection method
LOW (< 0.20)	Reproductive & fitness traits — conception rate, services per conception, calving interval, litter size, viability, egg number (0.2)	Family / progeny testing; improve by crossbreeding & management
MEDIUM (0.20–0.40)	Milk yield (0.25–0.30), growth rate, weaning weight, wool yield	Combined selection / pedigree + individual
HIGH (> 0.40)	Fat % & SNF % (0.5–0.6), egg weight (0.5), body/carcass measurements, mature body weight, fleece quality	Individual (mass) selection — the animal's own record is reliable

USES OF h^2 — SAY ALL FIVE

① Decides the **method of selection** ② **predicts genetic gain** $\Delta G = h^2 S = i \cdot h^2 \sigma_p$ ③ used in the **selection index** and in **BLUP** ④ estimates **breeding value** ($EBV = h^2 \times$ deviation) ⑤ tells how much superiority is actually **transmitted**. **Factors affecting h^2 :** trait, breed, variability, environment (a uniform environment raises h^2), method of estimation, inbreeding.

23 COMPONENTS OF VARIANCE & $G \times E$ ★ DRAW THIS TREE

$$P = G + E + G \times E \rightarrow V_P = V_G + V_E + V_{GE}$$

$$V_G = V_A + V_D + V_I$$

$$V_E = V_{Ep} + V_{Et}$$

V_A additive — **fixable, transmitted, the ONLY part selection uses**

V_{Ep} permanent environment (non-fixable but persists on the same animal)

V_D dominance (interaction **within** a locus) — exploited by **crossbreeding/heterosis**

V_{Et} temporary environment (season, lactation, feeding of the day)

V_I epistatic (interaction **between** loci) — also non-fixable

V_{GE} genotype $\times$ environment interaction

$G \times E$ INTERACTION = the **ranking or relative performance of genotypes changes from one environment to another**; i.e. the difference between two genotypes is not the same in all environments. Detected when **reaction norms cross**, or when the **genetic correlation of the same trait measured in two environments is well below 1** ($r_A < 0.8$ is taken as important $G \times E$).

THE CLASSIC INDIAN EXAMPLE — USE IT

Holstein-Friesian cattle rank first for milk yield in a **temperate, well-managed** environment but may rank **below Sahiwal or crossbreds** under **tropical heat, poor feeding and high disease challenge**. Hence: **select animals in the environment in which their progeny will actually be kept**; imported semen from temperate progeny tests may not perform here. Also seen in poultry strains under cage vs deep-litter, and in $G \times E$ for sheep in hill vs plain systems. Reducing $G \times E$: separate breeding programmes per production system, or select for **robustness/adaptability**.

Genotype–environment CORRELATION (different from interaction) = better genotypes systematically receive better environments (e.g. the best cows are given the best feed) — it **inflates apparent genetic differences** and must be removed by **contemporary comparison / random allotment**.

25 REPEATABILITY • CORRELATIONS • LAST-10-MINUTES SWEEP

REPEATABILITY (r) = the **correlation between REPEATED RECORDS of the SAME trait on the SAME individual** (successive lactations, successive clips, successive litters). It measures the fraction of variance due to **permanent differences** (genetic + permanent environmental). Estimated as the **intraclass correlation** from an ANOVA of repeated records. **Uses:** ① sets the **upper limit of h^2** ② decides **how many records are worth taking** ③ allows **early culling** on first-lactation records ④ used to compute **MPPA**. Values: milk yield ≈ 0.4 – 0.5 , fat % ≈ 0.6 , litter size ≈ 0.15 .

$$MPPA = \bar{P} + [nr / (1 + (n-1)r)] \times (\bar{P}_i - \bar{P})$$

GENETIC CORRELATION (r_A) = correlation between the **breeding values** of two different traits; caused by **PLEIOTROPY** (mainly) and by **linkage** (temporary). **PHENOTYPIC (r_p)** and **ENVIRONMENTAL (r_E)** correlations complete the set.

$$\text{Relation } r_p = r_A \cdot h_1 \cdot h_2 + r_E \cdot e_1 \cdot e_2$$

$$\text{Correlated response } CR_Y = i \cdot h_X \cdot h_Y \cdot r_A \cdot \sigma_{PY}$$

Uses of r_A : predicts the **correlated response** in a second trait • allows **indirect selection** when the target trait is hard/late/sex-limited to measure • essential for building a **selection index** • warns of **antagonistic pairs**. **Classic negative (antagonistic) r_A :** milk yield vs fat %, milk yield vs fertility, growth rate vs feed efficiency in poultry, egg number vs egg size. Positive: body weight and growth rate; milk yield and SNF yield.

Father of Genetics — Mendel; **coined "genetics"** — Bateson; **coined "gene"** — Johannsen

Cattle 2n = 60; buffalo 50; chicken 78; mule 63, sterile

Crossing over — pachytene; **chiasma** — diplotene; **synapsis** — zygotene

Barr bodies = $X - 1$; tortoiseshell tom = **XXY**

SRV on **Yp** = TDF; birds — **male ZZ, female ZW**

Freemartin — 92% of heifers twin-born with a bull; **XX/XY chimera**

1/29 Robertsonian translocation — cattle, ↓ fertility, screen AI bulls

Max RF = 50%; 1% RF = 1 cM; linkage groups = **n**

13:3 inhibitory • **15:1** duplicate dominant •

9:7 complementary • **12:3:1** dominant epistasis • **9:3:4** recessive epistasis • **2:1**

lethal • **Walnut:Rose:Pea:Single = 9:3:3:1**

HWE reached in **one** generation (autosomal); **2pq max = 0.5**

Systematic — mutation, migration, selection; **dispersive** — drift

Inbreeding changes genotypic, **not** gene, frequency

$h^2 = 4t$ (half-sib), **$2t$** (full-sib), **$2b$** (one parent), **R/S** (realised)

$r \geq h^2$; milk yield $h^2 \approx 0.25$, fat % ≈ 0.5 – 0.6

Only V_A is fixable by selection; **V_D** is

exploited by crossbreeding

PCR — Mullis 1983; 94 → 55 → 72 °C, **Taq**;

Southern DNA • **Northern RNA** • **Western**

protein

SUBJECTIVE ATTACK PLANS

"Heritability — definition, methods of estimation and uses" → definition (narrow & broad) → V_p tree → 5 methods with formulas → the 3-class value table → 5 uses → factors affecting it → conclusion linking to ΔG . "**Hardy–Weinberg law**" → statement → equation → 6 conditions → proof by random union of gametes (checkerboard) → attained in one generation → applications → χ^2 test → limitations. "**Components of variance**" → $P = G + E$ → the full tree → define each term → which is fixable → link to h^2 , heterosis and $G \times E$.